

First/contact author & main contributor

- [1] A. A. Vigliano and F. Longo. *Soccer Ball Tessellation: a novel method for tiling large uncertainty areas with narrow field of view telescopes*. In preparation. 2026.
Article based on the results of the PhD thesis and its new developments
- [2] A. A. Vigliano, F. Longo, and Ž. Bošnjak. “Reassessing high-energy emission correlations in gamma-ray bursts using a large, homogeneous sample of X-ray afterglows”. In: *Astron. Astrophys.* 710 (2026), A393. DOI: [10.1051/0004-6361/202558494](https://doi.org/10.1051/0004-6361/202558494). arXiv: [2605.25644](https://arxiv.org/abs/2605.25644) [astro-ph.HE].
Conceived the analysis, developed the methodology, performed the data analysis, wrote the paper
- [3] A. A. Vigliano. “Novel strategies for the observations of fast gamma-ray transients”. Publicly available [online](#). PhD thesis. Università degli Studi di Udine, Udine, Italy, 2025.
PhD thesis. I conceived the analysis, developed the methodology, performed the data analysis, wrote the paper
- [4] A. A. Vigliano and F. Longo. “Gamma-ray Bursts: 50 Years and Counting!” In: *Universe* 10.2 (2024), p. 57. DOI: [10.3390/universe10020057](https://doi.org/10.3390/universe10020057).
A review of the history of gamma-ray burst observations and studies. I conducted in-depth historical and scientific research and wrote the manuscript.
- [5] A. Kouchner, A. A. Vigliano, and M. Lamoureux. “Detection and phenomenology of cosmic neutrinos”. In: *PoS QG-MMSchools* (2024), p. 005. DOI: [10.22323/1.440.0005](https://doi.org/10.22323/1.440.0005).
A review of the history of observations and studies of astrophysical neutrinos as part of the Second School of the COST Action QGMM. I conducted extensive historical and scientific research and wrote the manuscript.
- [6] A. A. Vigliano et al. “Gamma Ray Burst localization with GECCO’s BGO anti-coincidence shields”. In: *PoS ICRC2023* (2023), p. 748. DOI: [10.22323/1.444.0748](https://doi.org/10.22323/1.444.0748).
Conceived the analysis, developed the methodology, performed the data analysis, wrote the paper

Contributor

- [7] R. Aloisio et al. “Crystal Eye: All sky MeV monitor with high precision real-time localization”. In: *Astroparticle Physics* 174 (2026), p. 103171. ISSN: 0927-6505. DOI: <https://doi.org/10.1016/j.astropartphys.2025.103171>. URL: <https://www.sciencedirect.com/science/article/pii/S0927650525000945>.
Contributed to the evaluation of the localization capabilities of transient events (especially GRBs) through classical and ML techniques
- [8] R. Alves Batista et al. “White paper and roadmap for quantum gravity phenomenology in the multi-messenger era”. In: *Class. Quant. Grav.* 42.3 (2025), p. 032001. DOI: [10.1088/1361-6382/ad605a](https://doi.org/10.1088/1361-6382/ad605a). arXiv: [2312.00409](https://arxiv.org/abs/2312.00409) [gr-qc].
Contributed to writing and review. Active member of the COST Action QGMM and co-proposer of its followup COST Action BridgeQG
- [9] A. Campoy-Ordaz et al. “Search for Lorentz Invariance Violation with time-lag on gamma-ray Cherenkov Telescope data : From the data to the time lag constraints”. In: *PoS QG-MMSchools* (2024), p. 009. DOI: [10.22323/1.440.0009](https://doi.org/10.22323/1.440.0009).
Contributed to the writing (tutorial section of the paper)
- [10] A. Coogan et al. “Hunting for dark matter and new physics with GECCO”. In: *Phys. Rev. D* 107.2 (2023), p. 023022. DOI: [10.1103/PhysRevD.107.023022](https://doi.org/10.1103/PhysRevD.107.023022). arXiv: [2101.10370](https://arxiv.org/abs/2101.10370) [astro-ph.HE].
Contributed with the evaluation of the detector’s capabilities
- [11] A. Donini et al. “Performance study update of observations in divergent mode for the Cherenkov Telescope Array”. In: *PoS ICRC2023* (2023), p. 840. DOI: [10.22323/1.444.0840](https://doi.org/10.22323/1.444.0840). arXiv: [2309.14106](https://arxiv.org/abs/2309.14106) [astro-ph.HE].
Contributed as an expert on tessellation strategies for the follow-up of poorly localized transient events

The publications listed below are large-scale collaborative works. I am member of both analysis and operations teams contributing to data taking, validation, and internal reviews.

Collaboration papers

- [12] G. Principe et al. “Broadband multiwavelength properties of the archetypal blazar 3C 279 during the 2017 Event Horizon Telescope campaign”. In: (June 2026). arXiv: [2606.25868 \[astro-ph.HE\]](#).
- [13] K. Abe et al. “Detection of the distant quasar OP 313 with the first Large-Sized Telescope of CTAO”. In: (May 2026). arXiv: [2605.26681 \[astro-ph.HE\]](#).
- [14] K. Abe et al. “Long-term study of the gamma-ray emission of Cygnus X-3 with MAGIC and Fermi-LAT”. In: (Apr. 2026). DOI: [10.1016/j.jheap.2026.100630](#). arXiv: [2604.27671 \[astro-ph.HE\]](#).
- [15] K. Abe et al. “The January 2010 flare of Mrk 421: Insights from a stochastic acceleration model”. In: *Astron. Astrophys.* 710 (2026), A201. DOI: [10.1051/0004-6361/202659090](#). arXiv: [2604.20480 \[astro-ph.HE\]](#).
- [16] S. Abe et al. “Chasing Gamma-Ray Signals from Binary Neutron Star Coalescences with the Cherenkov Telescope Array: Prospects and Observing Strategies”. In: *Astrophys. J.* 1004.1 (2026), p. 46. DOI: [10.3847/1538-4357/ae658c](#). arXiv: [2604.08748 \[astro-ph.HE\]](#).
- [17] K. Abe et al. “Extreme Blazars Observed with MAGIC: Second Catalog Release”. In: *Astrophys. J.* 1002.2 (2026), p. 138. DOI: [10.3847/1538-4357/ae5a8a](#). arXiv: [2604.05576 \[astro-ph.HE\]](#).
- [18] S. Abe et al. “MAGIC observations of NGC 4278. The first low-luminosity radio galaxy with compact jets detected at TeV energies”. In: (Mar. 2026). arXiv: [2603.22955 \[astro-ph.HE\]](#).
- [19] K. Abe et al. “Very-high-energy gamma-ray intranight variability from BL Lacertae during the extreme flaring state of 2022”. In: *Astron. Astrophys.* 710 (2026), A41. DOI: [10.1051/0004-6361/202557924](#). arXiv: [2603.19889 \[astro-ph.HE\]](#).
- [20] K. Abe et al. “First detection of VHE gamma-ray signal from the FSRQ TON 0599”. In: *Mon. Not. Roy. Astron. Soc.* 546 (2026), p. 4. DOI: [10.1093/mnras/stag032](#). arXiv: [2601.03873 \[astro-ph.HE\]](#).
- [21] J. Abhir et al. “Prompt Searches for Very-high-energy γ -Ray Counterparts to IceCube Astrophysical Neutrino Alerts”. In: *Astrophys. J.* 997.2 (2026), p. 141. DOI: [10.3847/1538-4357/ae2c4e](#). arXiv: [2512.16562 \[astro-ph.HE\]](#).
- [22] K. Abe et al. “CTAO LST-1 observations of magnetar SGR 1935+2154: Deep limits on sub-second bursts and persistent tera-electronvolt emission”. In: *Astron. Astrophys.* 706 (2026), A25. DOI: [10.1051/0004-6361/202555431](#). arXiv: [2512.05634 \[astro-ph.HE\]](#).
- [23] K. Abe et al. “VHE γ -ray observations of bright BL Lacs with the Large-Sized Telescope prototype (LST-1) of the CTAO”. In: *Mon. Not. Roy. Astron. Soc.* 544.1 (2025), pp. 669–686. DOI: [10.1093/mnras/staf1728](#). arXiv: [2510.03702 \[astro-ph.HE\]](#).
- [24] M. Tambone et al. “WINK: Advancing X and Gamma Ray Detection Technology for Space Observations”. In: *PoS ICRC2025* (2025), p. 857. DOI: [10.22323/1.501.0857](#).
- [25] A. Tarana et al. “X-ray Binaries: A Key Corollary Science with Crystal Eye”. In: *PoS ICRC2025* (2025), p. 594. DOI: [10.22323/1.501.0594](#).
- [26] K. Abe et al. “Time-dependent Modeling of the Subhour Spectral Evolution during the 2013 Outburst of Mrk 421”. In: *Astrophys. J.* 998.1 (2026), p. 6. DOI: [10.3847/1538-4357/ae25f0](#). arXiv: [2509.08686 \[astro-ph.HE\]](#).
- [27] K. Abe et al. “Constraining the TeV gamma-ray emission of SN 2024bch, a possible type II-L from a red supergiant progenitor - Multiwavelength observations and analysis of the progenitor”. In: *Astron. Astrophys.* 702 (2025), A125. DOI: [10.1051/0004-6361/202554721](#). arXiv: [2508.20048 \[astro-ph.HE\]](#).
- [28] K. Abe et al. “Prospects for dark matter observations in dwarf spheroidal galaxies with the Cherenkov Telescope Array Observatory”. In: *Mon. Not. Roy. Astron. Soc.* 544.3 (2025), pp. 2946–2986. DOI: [10.1093/mnras/staf1798](#). arXiv: [2508.19120 \[astro-ph.HE\]](#).
- [29] K. Abe et al. “Very high energy observations of the Seyfert galaxy NGC 4151 with MAGIC - Indication of another gamma-ray obscured candidate neutrino source”. In: *Astron. Astrophys.* 702 (2025), A38. DOI: [10.1051/0004-6361/202554723](#). arXiv: [2507.16527 \[astro-ph.HE\]](#).
- [30] S. Abe et al. “Testing the ubiquitous presence of very high energy emission in gamma-ray bursts with the MAGIC telescopes”. In: *Astron. Astrophys.* 700 (2025), A96. DOI: [10.1051/0004-6361/202555468](#). arXiv: [2507.05215 \[astro-ph.HE\]](#).

- [31] K. Abe et al. “GRB 221009A: Observations with LST-1 of CTAO and Implications for Structured Jets in Long Gamma-Ray Bursts”. In: *Astrophys. J. Lett.* 988.2 (2025), p. L42. DOI: [10.3847/2041-8213/ade4cf](https://doi.org/10.3847/2041-8213/ade4cf). arXiv: [2507.03077](https://arxiv.org/abs/2507.03077) [[astro-ph.HE](#)].
- [32] K. Abe et al. “Detection of the Geminga pulsar at energies down to 20 GeV with the LST-1 of CTAO”. In: *Astron. Astrophys.* 698 (2025), A283. DOI: [10.1051/0004-6361/202554350](https://doi.org/10.1051/0004-6361/202554350). arXiv: [2505.21632](https://arxiv.org/abs/2505.21632) [[astro-ph.HE](#)].
- [33] K. Abe et al. “Very high-energy gamma-ray detection and long-term multiwavelength view of the flaring blazar B2 1811+31”. In: *Astron. Astrophys.* 697 (2025), A172. DOI: [10.1051/0004-6361/202452942](https://doi.org/10.1051/0004-6361/202452942). arXiv: [2503.19054](https://arxiv.org/abs/2503.19054) [[astro-ph.HE](#)].
- [34] K. Abe et al. “Detection of RS Oph with LST-1 and modelling of its HE/VHE gamma-ray emission”. In: *Astron. Astrophys.* 695 (2025), A152. DOI: [10.1051/0004-6361/202452447](https://doi.org/10.1051/0004-6361/202452447). arXiv: [2503.13283](https://arxiv.org/abs/2503.13283) [[astro-ph.HE](#)].
- [35] K. Abe et al. “Insights from the first flaring activity of a high synchrotron peaked blazar with X-ray polarization and VHE gamma rays”. In: *Astron. Astrophys.* 695 (2025), A217. DOI: [10.1051/0004-6361/202452785](https://doi.org/10.1051/0004-6361/202452785). arXiv: [2410.23140](https://arxiv.org/abs/2410.23140) [[astro-ph.HE](#)].
- [36] K. Abe et al. “A new method of reconstructing images of gamma-ray telescopes applied to the LST-1 of CTAO”. In: *Astron. Astrophys.* 691 (2024), A328. DOI: [10.1051/0004-6361/202450889](https://doi.org/10.1051/0004-6361/202450889). arXiv: [2410.16042](https://arxiv.org/abs/2410.16042) [[astro-ph.HE](#)].
- [37] K. Abe et al. “A detailed study of the very high-energy Crab pulsar emission with the LST-1”. In: *Astron. Astrophys.* 690 (2024), A167. DOI: [10.1051/0004-6361/202450059](https://doi.org/10.1051/0004-6361/202450059). arXiv: [2407.02343](https://arxiv.org/abs/2407.02343) [[astro-ph.HE](#)].
- [38] K. Abe et al. “Galactic transient sources with the Cherenkov Telescope Array Observatory”. In: *Mon. Not. Roy. Astron. Soc.* 540.1 (2025), pp. 205–238. DOI: [10.1093/mnras/staf655](https://doi.org/10.1093/mnras/staf655). arXiv: [2405.04469](https://arxiv.org/abs/2405.04469) [[astro-ph.HE](#)].
- [39] K. Abe et al. “GRB 221009A observations with LST-1 at VHE gamma rays”. In: *PoS HEPROVIII* (2024), p. 058. DOI: [10.22323/1.461.0058](https://doi.org/10.22323/1.461.0058).
- [40] S. Abe et al. “Dark matter line searches with the Cherenkov Telescope Array”. In: *JCAP* 07 (2024), p. 047. DOI: [10.1088/1475-7516/2024/07/047](https://doi.org/10.1088/1475-7516/2024/07/047). arXiv: [2403.04857](https://arxiv.org/abs/2403.04857) [[hep-ph](#)].
- [41] K. Abe et al. “Star tracking for pointing determination of Imaging Atmospheric Cherenkov Telescopes - Application to the Large-Sized Telescope of the Cherenkov Telescope Array”. In: *Astron. Astrophys.* 679 (2023), A90. DOI: [10.1051/0004-6361/202347128](https://doi.org/10.1051/0004-6361/202347128).
- [42] S. Abe et al. “Prospects for a survey of the galactic plane with the Cherenkov Telescope Array”. In: *JCAP* 10 (2024), p. 081. DOI: [10.1088/1475-7516/2024/10/081](https://doi.org/10.1088/1475-7516/2024/10/081). arXiv: [2310.02828](https://arxiv.org/abs/2310.02828) [[astro-ph.HE](#)].
- [43] H. Abe et al. “Performance of the joint LST-1 and MAGIC observations evaluated with Crab Nebula data”. In: *Astron. Astrophys.* 680 (2023), A66. DOI: [10.1051/0004-6361/202346927](https://doi.org/10.1051/0004-6361/202346927). arXiv: [2310.01954](https://arxiv.org/abs/2310.01954) [[astro-ph.IM](#)].
- [44] K. Abe et al. “Prospects for γ -ray observations of the Perseus galaxy cluster with the Cherenkov Telescope Array”. In: *JCAP* 10 (2024), p. 004. DOI: [10.1088/1475-7516/2024/10/004](https://doi.org/10.1088/1475-7516/2024/10/004). arXiv: [2309.03712](https://arxiv.org/abs/2309.03712) [[astro-ph.HE](#)].
- [45] H. Abe et al. “Joint Observation of the Galactic Center with MAGIC and CTA-LST-1”. In: *PoS ICRC2023* (2023), p. 573. DOI: [10.22323/1.444.0573](https://doi.org/10.22323/1.444.0573).
- [46] K. Abe et al. “Galactic Center Studies with CTA-LST-1”. In: *PoS ICRC2023* (2023), p. 574. DOI: [10.22323/1.444.0574](https://doi.org/10.22323/1.444.0574).
- [47] K. Abe et al. “First results of pulsar observations with the LST-1”. In: *PoS ICRC2023* (2023), p. 569. DOI: [10.22323/1.444.0569](https://doi.org/10.22323/1.444.0569).
- [48] K. Abe et al. “Imaging Atmospheric Cherenkov Telescopes pointing determination using the trajectories of the stars in the field of view”. In: *PoS ICRC2023* (2023), p. 566. DOI: [10.22323/1.444.0566](https://doi.org/10.22323/1.444.0566). arXiv: [2310.04101](https://arxiv.org/abs/2310.04101) [[astro-ph.IM](#)].
- [49] K. Abe et al. “Chasing Gravitational Waves with the Cherenkov Telescope Array”. In: *PoS ICRC2023* (2023), p. 1534. DOI: [10.22323/1.444.1534](https://doi.org/10.22323/1.444.1534). arXiv: [2310.07413](https://arxiv.org/abs/2310.07413) [[astro-ph.HE](#)].

- [50] R. M. Dominik et al. “Interpolation of Instrument Response Functions for the Cherenkov Telescope Array in the Context of pyirf”. In: *PoS ICRC2023* (2023), p. 618. DOI: [10.22323/1.444.0618](https://doi.org/10.22323/1.444.0618). arXiv: [2309.16488](https://arxiv.org/abs/2309.16488) [[astro-ph.IM](#)].
- [51] K. Abe et al. “Characterization of SiPM and development of test bench modules for the next-generation cameras for Large-Sized Telescopes for Cherenkov Telescope Array”. In: *PoS ICRC2023* (2023), p. 635. DOI: [10.22323/1.444.0635](https://doi.org/10.22323/1.444.0635). arXiv: [2309.10411](https://arxiv.org/abs/2309.10411) [[astro-ph.IM](#)].
- [52] K. Abe et al. “Detection of the 2021 Outburst of RS Ophiuchi with the LST-1”. In: *PoS ICRC2023* (2023), p. 677. DOI: [10.22323/1.444.0677](https://doi.org/10.22323/1.444.0677). arXiv: [2310.09683](https://arxiv.org/abs/2310.09683) [[astro-ph.HE](#)].
- [53] M. Linhoff et al. “Performance update of an event-type based analysis for the Cherenkov Telescope Array”. In: *PoS ICRC2023* (2023), p. 738. DOI: [10.22323/1.444.0738](https://doi.org/10.22323/1.444.0738). arXiv: [2309.11375](https://arxiv.org/abs/2309.11375) [[astro-ph.IM](#)].
- [54] H. Abe et al. “Low energy performance boost through a hardware stereoscopic trigger between CTA LST1 and MAGIC”. In: *PoS ICRC2023* (2023), p. 700. DOI: [10.22323/1.444.0700](https://doi.org/10.22323/1.444.0700).
- [55] M. C. Strzys et al. “Pybkgmodel - a background modelling toolbox for the CTA”. In: *PoS ICRC2023* (2023), p. 894. DOI: [10.22323/1.444.0894](https://doi.org/10.22323/1.444.0894).
- [56] K. Abe et al. “First Science Results from CTA LST-1 Telescope and status of LST2-4”. In: *PoS ICRC2023* (2023), p. 731. DOI: [10.22323/1.444.0731](https://doi.org/10.22323/1.444.0731).
- [57] K. Abe et al. “Multi-wavelength analysis of the PeVatron candidate LHAASO J2108+5157”. In: *PoS ICRC2023* (2023), p. 727. DOI: [10.22323/1.444.0727](https://doi.org/10.22323/1.444.0727).
- [58] F. G. Saturni et al. “Dark matter searches in dwarf spheroidal galaxies with the Cherenkov Telescope Array”. In: *PoS ICRC2023* (2023), p. 1366. DOI: [10.22323/1.444.1366](https://doi.org/10.22323/1.444.1366). arXiv: [2309.09607](https://arxiv.org/abs/2309.09607) [[astro-ph.HE](#)].
- [59] H. Abe et al. “Observations of the Crab Nebula and Pulsar with the Large-sized Telescope Prototype of the Cherenkov Telescope Array”. In: *Astrophys. J.* 956.2 (2023), p. 80. DOI: [10.3847/1538-4357/ace89d](https://doi.org/10.3847/1538-4357/ace89d). arXiv: [2306.12960](https://arxiv.org/abs/2306.12960) [[astro-ph.HE](#)].
- [60] H. Abe et al. “RS Ophiuchi nova outburst detection by the LST-1”. In: *PoS Gamma2022* (2023), p. 055. DOI: [10.22323/1.417.0055](https://doi.org/10.22323/1.417.0055).
- [61] H. Abe et al. “Observation of the galactic PeVatron candidate LHAASO J2108+5157 with the Large-Sized Telescope for the Cherenkov Telescope Array”. In: *PoS Gamma2022* (2023), p. 129. DOI: [10.22323/1.417.0129](https://doi.org/10.22323/1.417.0129). arXiv: [2212.02123](https://arxiv.org/abs/2212.02123) [[astro-ph.HE](#)].
- [62] S. Abe et al. “Multi-wavelength study of the galactic PeVatron candidate LHAASO J2108+5157”. In: *Astron. Astrophys.* 673 (2023), A75. DOI: [10.1051/0004-6361/202245086](https://doi.org/10.1051/0004-6361/202245086). arXiv: [2210.00775](https://arxiv.org/abs/2210.00775) [[astro-ph.HE](#)].